VEPOLINK
MONITORING PURITY

ENVIRONMENTAL INTELLIGENCE × GAS ANALYSIS

CONTINUOUS AMBIENT AIR QUALITY MONITORING

Ambient CO_2 Gas Analyzer *NDIR Gas Filter Correlation*

Carbon dioxide measured across dual ranges with a long-path curved mirror cell and a durable gas filter correlation wheel — with automatic zero correction and a year of logged data on board.



0–5 · 0–30 %
DUAL MEASURING CHANNELS

7 inch
TFT LCD TOUCHSCREEN

1 year
ON-BOARD DATA LOGGING

NDIR GFC REFERENCE METHOD | **DUAL CHANNEL** 0-5% / 0-30% | **CPCB** ALIGNED REPORTING

<0.2ppm
LOWER DETECTABLE LIMIT

2
NDIR CHANNELS, DUAL RANGE

<30s
RESPONSE TIME T63

32bit
ARM PROCESSOR, LINUX BASED

01 OVERVIEW

Carbon dioxide, *across two ranges at once.*

A continuous ambient CO_2 analyzer built on NDIR gas filter correlation, with dual sensor channels covering 0–5% and 0–30% CO_2 from one instrument. A curved mirror gives a long optical path in a compact cell, and a no-leak GFC wheel keeps the signal stable over a long service life.

A 32-bit ARM processor runs on Linux with a 7-inch touchscreen, automatic zero correction, and Ethernet, RS-232, RS-485, USB and analog interfaces — reporting to a station, SCADA and a regulatory server at once.

REPORTED CHANNELS

- CO_2 CHANNEL A (0-5%)
- CO_2 CHANNEL B (0-30%)
- 1-MIN AVERAGE
- HOURLY AVERAGE
- 24-HR AVERAGE
- SAMPLE FLOW
- CELL TEMPERATURE
- IR SOURCE INTENSITY
- ZERO & SPAN STATUS
- FAULT DIAGNOSTICS

· PRODUCT FEATURES

Stable Infrared Source

An IR source selected for stable output keeps the correlation signal steady between calibrations and extends service intervals.

Dual-Range Sensing

Two NDIR channels — 0–5% and 0–30% CO_2 — cover ambient background and elevated concentrations from a single, shared sample path.

Touchscreen Operation

A 7-inch TFT LCD touch interface puts live values, trends, ranges and diagnostics in front of the operator.

One Year of Data On Board

An internal NAND-flash logger holds up to a year of one-minute averages, downloadable over USB or network.

02 MEASUREMENT PRINCIPLE

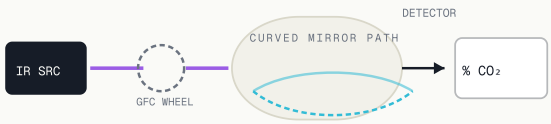
How NDIR GFC *counts CO_2.*

Carbon dioxide absorbs infrared light at a specific wavelength. Comparing absorption through a spinning gas filter correlation wheel against a reference gives a signal proportional to concentration — a direct optical measurement with no consumables in the sample path.

NDIR

Long-Path Absorption

IR SOURCE · CURVED MIRROR CELL · DETECTOR



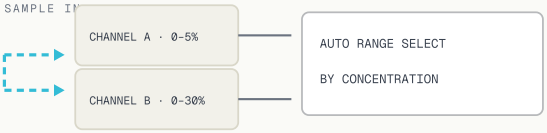
Infrared light travels a folded, long optical path through a curved-mirror cell holding the sample. CO_2 absorbs a characteristic wavelength; the detector reads the resulting attenuation.

`Absorption ∝ CO₂ concentration`

DUAL RANGE

Two NDIR Channels

0-5% CHANNEL · 0-30% CHANNEL



The shared sample splits across two independent NDIR sensor channels, one tuned for background ambient CO_2 and one for elevated concentrations, so both ranges report from one instrument.

`One sample, two ranges`

STABILITY

Auto Zero & Compensation

AUTOMATIC BASELINE · T AND P CORRECTION



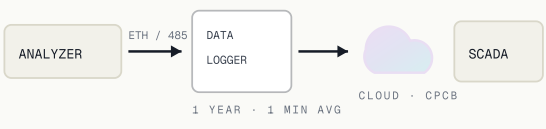
An automatic zero correction cycle re-establishes the baseline without operator intervention, while temperature and pressure compensation holds the reading steady as ambient conditions move through the day.

`Unattended long-term operation`

DATA PATH

Logging & Reporting

ETHERNET · RS-232 / 485 · USB · ANALOG



Readings are stored internally as one-minute averages for up to a year and transmitted live to a station data logger, plant SCADA or a pollution control board server.

`Download over USB or network`

03 SAMPLE PATH

From inlet to *reading.*

Everything between the sample line and the display is designed to keep flow constant, the optical path clean, and both channels reporting the same instant of air.

STEP 01

Filtered intake

Ambient air passes a 47 mm Teflon particulate filter and inert sample line into the analyzer.

STEP 02

Critical-orifice flow

A critical orifice stabilises sample flow at 800 cc/min ±10%, so both NDIR channels see a consistent draw.

STEP 03

Dual NDIR cells

The sample fills two curved-mirror cells, each paired with a GFC wheel tuned to its own range.

STEP 04

Detect and report

Both channels are corrected for temperature and pressure, then displayed, logged and transmitted.

WHY DUAL CHANNELS MATTER

Ambient background CO_2 sits in the low hundreds of ppm, while a plant vent, greenhouse or confined space can run into the percent range. A single fixed range forces a trade-off between resolution and headroom; two independent NDIR channels give resolution at background levels and headroom at elevated levels, from the same sample draw.

FLOW CONTROL

A critical orifice holds sample flow to 800 cc/min ±10% regardless of small pressure changes upstream, so the residence time in each optical cell — and therefore the calibration — stays consistent between sampling events.

WHY CARBON DIOXIDE IS MONITORED

CO_2 is the principal long-lived greenhouse gas and, at high indoor or confined-space concentrations, a direct occupant-safety hazard through displacement of oxygen. Ambient background tracking supports climate and air-quality reporting, while elevated-range monitoring covers process vents, greenhouses, fermentation and confined spaces where CO_2 can build up quickly. Continuous, dual-range measurement is what lets one instrument serve both needs.

04 SPECIFICATIONS

Technical *specifications.*

MEASUREMENT

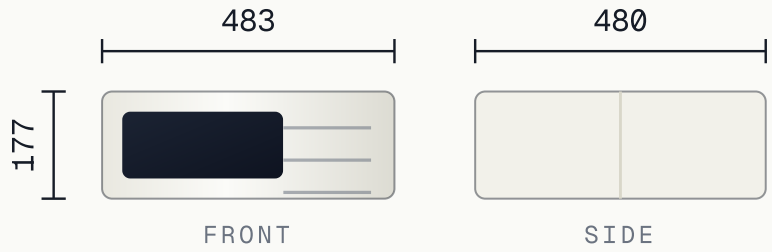
Parameter	Ambient carbon dioxide, CO_2
Working Principle	NDIR gas filter correlation
Measuring Ranges	0 - 5% / 0 - 30% CO_2, dual channel
Lower Detectable Limit	< 0.2 ppm
Accuracy	±(0.2% CO_2 + 3% of reading)
Repeatability	Within ±0.5% of span gas concentration
Linearity	Within ±1% of full scale
Precision	1% of reading or 0.1 ppm
Zero Drift	< 0.25 ppm
Span Drift	Within ±1% FS per day
Response Time	T63 < 30 s for ambient changes

SYSTEM · COMMS · BUILD

Display	7-inch graphic TFT LCD with touch interface
Controller	32-bit ARM Cortex-M4, Linux based multiprocessing
Zero Correction	Automatic · internal zero optional
Compensation	Temperature and pressure
Sample Flow	800 cc/min ±10%, critical orifice controlled
Particulate Filter	47 mm Teflon
Analog Output	0-1 V / 0-5 V or 4-20 mA (optional)
Data Logger	NAND flash · up to 1 year of 1-minute averages
Interfaces	Ethernet · USB 2.0 · RS-232 · RS-485
Power Supply	110 - 240 VAC · 50/60 Hz
Operating Conditions	5 - 40 °C operating · 0 - 50 °C, 0 - 95% RH non-condensing

INSTRUMENT DIMENSIONS

INDICATIVE FRONT & SIDE ELEVATION · MM



05 INSTALLATION & UPKEEP

Racked, calibrated, *left running.*

The analyzer is installed in a station cabinet or process rack with the sample manifold, calibration source and data logger. Sample flow is held at **800 cc/min ±10%** by a critical orifice, with a scheduled **zero and span check** traceable to certified gas across both ranges.

01 · MOUNTING

19-inch rack or bench

Standard rack fitment alongside other CAAQMS analyzers, sharing the station's manifold, power and data logger.

02 · SAMPLE LINE

Inert and short

PTFE line with the 47 mm Teflon particulate filter, kept short so CO_2 reaches both optical cells without loss.

03 · ZERO AIR

Clean baseline source

A zero air unit supplies scrubbed, CO_2-free air for the automatic zero cycle and for periodic manual verification.

04 · CALIBRATION

Two-range traceable

Span each channel with certified CO_2 gas appropriate to its range, via a multipoint calibrator, on the site's audit schedule.

05 · CONSUMABLES

Filter and orifice

Routine upkeep is the inlet filter, critical orifice inspection and mirror/optical cell check. No reagents and no wet chemistry.

06 · DIAGNOSTICS

Reported with the data

IR source intensity, cell temperature and pressure, flow and zero/span status travel alongside every reading.

PART OF A FULL STATION

The CO_2 analyzer is normally specified alongside CO, SO_2 and NOx analyzers, a multipoint calibrator and zero air unit, or, for elevated-range use, as a standalone process or confined-space monitor. Sharing a time base and data path with a station lets a background CAAQMS site and a process vent both report from one platform.

COMMISSIONING & AUDIT

Commissioning covers leak and flow verification, a linearity check across each channel's working range, and a documented zero and span baseline that becomes the reference for subsequent audits. Thereafter the automatic zero cycle carries day-to-day stability and the scheduled span check confirms it against certified gas, so the record stays defensible without a technician on site.

06 APPLICATIONS

Where it *works.*

From background climate tracking to process and confined-space safety, dual-range CO_2 measurement covers both ends of the concentration scale from one analyzer.

Regulatory CAAQMS Networks	Thermal Power Stations	Refineries & Petrochemicals
Chemical & Fertiliser Plants	Greenhouses & Fermentation	Confined Space Safety
Ports & Marine Fuel Zones	EIA & Baseline Studies	Research & Climate Studies

WHAT THE DATA SUPPORTS

Continuous, dual-range CO_2 records serve two very different audiences: climate and air-quality reporting, which needs stable, high-resolution background readings in the low-5% range, and process and safety monitoring, which needs headroom into the 30% range without switching instruments. One measurement, two conversations.

CHOOSING A CHANNEL

The 0–5% channel suits ambient background, greenhouse and general HVAC monitoring where resolution at typical levels matters most. The 0–30% channel suits process vents, fermentation headspace and confined-space entry checks where concentrations can rise sharply and headroom, not resolution, is the priority.

07 FAQ

Frequently asked *questions.*

Q What is the working principle?

Non-dispersive infrared gas filter correlation (NDIR GFC). A curved-mirror long-path cell and a spinning correlation wheel give a self-referencing signal proportional to CO_2 concentration.

Q Can it measure low CO_2 concentrations accurately?

Yes. The 0–5% channel is specifically designed for accurate, reliable measurement at background ambient levels, with a lower detectable limit under 0.2 ppm.

Q What are the two measuring ranges for?

The 0–5% channel covers ambient background monitoring; the 0–30% channel covers elevated concentrations from process vents, greenhouses or confined spaces — both from one analyzer.

Q Does it support automatic calibration?

Yes, an automatic zero correction (Auto Zero) system holds a stable baseline on both channels without operator intervention.

Q Is it suitable for continuous monitoring?

Yes. A stable IR source, Linux-based control and a robust build make it suitable for continuous, unattended operation.

Q What interfaces are available?

Ethernet, USB 2.0, RS-232, RS-485, and an optional analog output (0–1 V, 0–5 V or 4–20 mA) for flexible system integration.

Q How is sample flow controlled?

A critical orifice stabilises flow at 800 cc/min ±10%, keeping the residence time in each optical cell — and the calibration — consistent.

Q How much data does it hold?

The internal NAND-flash logger stores up to one year of one-minute averaged data on both channels, exportable over USB or the network.

Q Can it be installed outdoors?

Yes, housed in a weather-protected station cabinet. The analyzer itself is a rack instrument; the cabinet provides environmental protection and shared services.

Q What maintenance does it need?

Inlet filter changes, periodic leak and flow checks, and scheduled zero and span verification with certified gas on each channel. There are no reagents or wet chemistry.

This brochure is general product information. Configuration, dimensions and performance figures are indicative and confirmed at order. For installation guidelines and full technical detail, refer to the user manual supplied with the product. Specifications may change without notice as the product is improved.

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CARBON DIOXIDE, MEASURED CONTINUOUSLY

Know your CO_2. *In real time.*

Tell us the site, the range you expect and who receives the data, and we'll configure the analyzer and the station around it — calibration, zero air, cabinet and reporting. Pick how you'd like to start.

01 **Talk to our engineers**
Speak directly to the team that specifies, installs and calibrates ambient gas analyzers. No call centre.
+91 99538 86178 →

02 **Request a station configuration**
RECOMMENDED
Send your site and reporting requirement — we'll spec ranges, calibration, zero air, cabinet and communications.
sales@vepolink.com →

03 **Book a demonstration**
See a running analyzer and the reporting dashboard with our engineering team.
Request a demo →

WWW.VEPOLINK.COM · SALES@VEPOLINK.COM